Vocabulary Builder — Full Feature Analysis

Learning Engine

Spaced Repetition System (SM-2 Algorithm)

This is the same algorithm used by Anki, widely considered the gold standard in memory science. The app tracks three per-word variables: **ease factor** (starts at 2.5), **review interval** (days until next review), and **review count**. After each flashcard review, the algorithm adjusts the next review date based on how well the user recalled the word. Poor recall resets the interval; strong recall extends it exponentially. This is a proven technique from cognitive science that optimizes long-term retention while minimizing study time.

Three Mastery Levels per Word

- *Learning* — default for new words
- *Reviewing* — reached after 3+ reviews with ≥70% accuracy
- *Mastered* — reached after 10+ reviews with ≥90% accuracy and a ≥30 day review interval

Flashcard Mode

Front/back card review with audio pronunciation (fetched from the dictionary API) and example sentences. Integrated directly with the SM-2 engine so every flip updates that word's review schedule.

Quiz Mode

Multiple-choice format. Up to 10 words per session drawn from the user's active list. Shows the definition, user selects the correct word from 4 options. Tracks score and percentage at the end.

Daily Streak Tracking

Day-over-day consistency is measured and displayed. The streak increments for consecutive calendar days of activity and resets if a day is missed — a proven motivational mechanic.

Word & Content Management

Manual Word Entry

- Word, definition, example sentence, and custom tags — all user-controlled
- Words can be edited or deleted at any time

Automatic Dictionary Lookup

When a user adds a word, the app queries a dictionary API (proxied through the backend) to auto-populate definitions, example sentences, phonetic pronunciation, and audio. The Datamuse API is also queried for synonyms and antonyms. Users can also manually trigger the "Word Lookup" view to search any word before adding it.

CSV Bulk Import

Users upload a file with columns: word, definition, sentence. The app:

- Auto-detects whether a header row is present
- Recognizes common header variations ("term", "vocab", "meaning", "phrase", etc.)
- Skips duplicates (already in the user's list)

- Skips rows missing word or definition
- Shows a preview with counts (valid, duplicates, invalid) before committing
- Lets users apply a tag to the entire import batch

This is the key feature for businesses who want to load industry-specific word lists.

Search & Filter
Live search across word and definition fields. Filterable by mastery level and tags.

Tag System
Custom tags per word (or per import batch) for organizing vocabulary by topic, unit, or source.

Data & Sync

Cloud Storage — Firebase Firestore
Every word and its entire review history is stored in Firebase's real-time NoSQL cloud database, tied to the authenticated user's UID. This means:

- Words are synced in real time across every device the user logs into
- No manual backup needed
- Data persists if the user reinstalls the app or switches phones/computers

Subscriber Registry — PostgreSQL
A separate server-side database (PostgreSQL) maintains subscription status, trial dates, admin grants, and payment records. This is used by the admin panel and survives server restarts and redeployments — unlike file-based storage that some competing apps use.

Authentication

- **Email + Password** signup and login
- **Password reset** via email
- **Anonymous sign-in** for the direct checkout flow (no friction for purchasers)
- Session persistence — users stay logged in across browser sessions

---

Payments & Business Model

Pricing
500,000 ₫ / 6 months (≈$19 USD) + 10% VAT for Vietnamese purchasers. Total: 550,000 ₫.

7-Day Free Trial — standard access flow. Full features from day one.

Direct Subscription — "Skip Trial" flow for businesses and employees who want to pay immediately without a trial.

Paddle (Global Payment Processor)
Handles credit/debit cards, Apple Pay, Google Pay, and PayPal. Paddle manages tax compliance, recurring billing, cancellations, and refunds globally.

VietQR / Techcombank
Local Vietnamese bank transfer option. The user enters their email, the QR code is revealed, they transfer via any Vietnamese banking app, and the admin activates the account. This is critical for the Vietnamese B2B market where many businesses still prefer bank transfer over card.

UI & User Experience

Mobile-First Design
Built specifically to feel like a native mobile app in the browser. Vertical navigation, touch-friendly card sizes, 3D button press effects.

Dark Mode
Full dark/light mode toggle with preference saved to the browser. Every color, border, and gradient is theme-aware.

Responsive Layout
Works on phone, tablet, and desktop browsers without a separate mobile app.

No App Store Required
Runs entirely in the browser — no installation, no App Store approval, no update process for the user.

Progressive Loading
Splash screen while Firebase initializes. "Checking access..." screen while subscription loads. Prevents blank/broken states.

Security

- Helmet.js HTTP security headers
- API rate limiting
- Webhook signature verification (Paddle webhooks are validated before processing)
- Firebase Security Rules restrict each user's data to their own UID
- Admin password stored as an environment secret, never hardcoded

---

Where This App Stands vs. Competitors

Feature	Vocabulary Builder	Anki	Quizlet	Duolingo	Memrise
SM-2 Spaced Repetition	✅ Full	✅ Full	❌ Basic	❌ Gamified only	✅ Partial
CSV Industry Word Import	✅ Yes	✅ (complex)	✅ (paid tier)	❌	❌
Auto Dictionary + Audio	✅ Yes	❌ Manual	✅ Limited	✅ Built-in	✅ Limited
Synonyms / Antonyms	✅ Yes	❌	❌	❌	❌
Cross-Device Cloud Sync	✅ Realtime	⚠️ AnkiWeb (paid)	✅ Yes	✅ Yes	✅ Yes
Works in Browser (no install)	✅ Yes	❌ App required	✅ Yes	✅ Yes	✅ Yes
Dark Mode	✅ Yes	✅ Yes	✅ Yes	❌	❌
Vietnamese Bank Payment	✅ VietQR	❌	❌	❌	❌
Admin Override / B2B	✅ Full panel	❌	❌	❌	❌

Industry-Specific Focus	✅ Designed for it	⚠️ General	⚠️ General	❌ Consumer only	⚠️ General
One-Time Bulk Team Setup	✅ CSV batch	⚠️ Manual	⚠️ Limited	❌	❌
Price (6 months)	~$19	Free (app) / $25/yr (sync)	$35.99/yr	$79.99/yr	$59.99/yr

---

The Core Competitive Argument

Against Anki: Anki requires a separate desktop app and AnkiWeb subscription for sync. It has a steep learning curve and no auto-dictionary lookup. Vocabulary Builder offers the same SM-2 algorithm in a clean browser interface with automatic definitions, synonyms, and audio — at a lower cost.

Against Quizlet: Quizlet's free tier is increasingly limited, and its "spaced repetition" is basic compared to true SM-2. CSV import is only on paid plans. Vocabulary Builder offers deeper SRS, automatic word enrichment, and is purpose-built for professional vocabulary — not just student flashcards.

Against Duolingo/Babbel: Those are designed for complete language beginners following a fixed curriculum. They cannot be customized with your own industry vocabulary. Vocabulary Builder is for professionals who already know conversational English and need to master the specific vocabulary of their field.

Against generic apps in Vietnam: No other app combines SM-2 SRS + industry CSV import + auto-dictionary lookup + VietQR payment + a Vietnamese B2B admin console in one product.